Fractional indices

1

a $\sqrt[3]{x}$ b $\left(\sqrt{a}\right)^3$ c $\left(\sqrt[4]{20}\right)^5$ d $\dfrac{1}{\sqrt{y}}$

e $3\sqrt{x}$ f $\left(\sqrt[3]{b}\right)^7$ g $\left(\sqrt{y}\right)^3$ h $\dfrac{1}{\left(\sqrt[3]{5}\right)^4}$

2

a $2^{\frac{1}{3}}$ b $2^{\frac{4}{5}}$ c $2^{\frac{5}{2}}$ d $2^{\frac{1}{3}}$

e $2^{\frac{10}{3}}$ f $2^{a+\frac{1}{2}}$ g $2^{\frac{5}{2}}$ h $2^{-\frac{3}{2}}$

3

a $3^{\frac{3}{2}}$ b $3^{x+\frac{1}{4}}$ c $3^{\frac{5}{2}}$ d $3^{-\frac{2x}{3}}$

4

$$\begin{aligned}\sqrt{m^8} &= \left(m^8\right)^{\frac{1}{2}}\\ &= m^{8\times\frac{1}{2}}\\ &= m^4\end{aligned}$$

5

a $x^{\frac{1}{2}}$ b $x^{\frac{1}{6}}$ c $x^{-\frac{1}{2}}$ d $x^{\frac{2}{3}}$

e $x^{\frac{3}{4}}$ f $x^{\frac{6}{7}}$ g x^2 h $x^{-\frac{1}{2}}$

i $x^{\frac{1}{4}}$ j x^{2a} k $x^{\frac{3}{2}}$ l $x^{\frac{3}{2}}$

m $x^{-\frac{5}{4}}$ n $x^{\frac{1}{10}}$ o $x^{\frac{7}{3}}$

6

a $64^{\frac{2}{3}} = \left(\sqrt[3]{64}\right)^2$ b 16

7

a 1 b 8 c 11 d 3

e 5 f -4 g $\dfrac{1}{2}$ h $\dfrac{1}{27}$

i -4 j $\dfrac{3}{10}$ k 8 l $\dfrac{4}{25}$

m 16 n $\dfrac{1}{4}$ o $-\dfrac{1}{13}$ p $-\dfrac{4}{5}$

q 2 r -2 s 10 t 3

8

a 31.62 b 10.08 c 3.34 d 62.90

9

a b^4 b m^3 c $a^{\frac{5}{4}}$ d $a^{-\frac{6}{5}}$

e x^2 f m g jx^2 h $x^{30}y^{18}$

i $2a^4$ j $25u^8v^6$ k $4b^{\frac{1}{12}}$ l $y^{\frac{10}{3}}$

m $2x+9$ n $9x^6y^4$ o $\dfrac{x^6}{16}$ p $\dfrac{6x^9}{y^{10}}$

q $4y+3$ r $y\left(x+9\right)$

10

a $\sqrt{x+3}$ b $\sqrt[3]{x+5}$

11 One way of describing the expression is to say that we are raising m to the power of q, then taking the rth root of the result. Alternatively, we can describe the expression as taking the rth root of m, and then raising the result to the power of q.

12

a No b Yes c No d Yes

13 No. An even power of any real number is non-negative. As -16 is negative, it cannot have a real valued fourth root.

14

a $m^{\frac{13}{2}}$ b $\sqrt{m^{13}}$

15

a $m^{\frac{1}{4}} = (m^{\frac{1}{2}})^{\frac{1}{2}}$
$= (\sqrt{m})^{\frac{1}{2}}$

b $m = 64$

16

a $\dfrac{3}{2}$ b $x = 27$

17

a 2 b 125 c 4 d 27

18

a $x = 625$ b $x = 20$ c $x = 14$ d $x = -\dfrac{11}{16}$

19 $k = 6$

20 It will be easier to use the property $a^{\frac{m}{n}} = ($[illegible], finding the square root of 81 (an easily identifiable square number) first and then find the cube of the result. Rather than be required to cube 81 (a large number) and then find the square root of the result.

Applications

21

a $A = 24 \text{ cm}^2$

b $V = 216 \text{ cm}^3$

22

a $V = 33 \text{ cm}^3$

b $A = 121 \text{ cm}^2$

23

a $V = \dfrac{2\sqrt{2}}{3} \text{ cm}^3$

b $a = \sqrt{2}\,(3V)^{\frac{1}{3}}$

c $a = 6\sqrt{2} \text{ cm}$

24

a $\$700$

b $n = 1296$

25

a $-\$2560$

b $n = 11\,434$